

InteractAI

Build sheet — P0 only

Brett Cooper | 24-day window | condensed from spec v1.0

1 What you are building

A real-time voice simulator for conversations that are hard to practise alone. The user picks a scenario — a backend screening, a salary negotiation, a viva — and speaks. An AI persona answers out loud, in character, in under one and a half seconds, and keeps the pressure on: it follows up on vague answers, paces itself against the clock, and never coaches. Behind the conversation a second agent scores every turn against an explicit rubric and returns a report that quotes the user’s own words back with timestamps and playable audio.

The persona sits on the latency critical path and is optimised for speed. The coach sits off it and is optimised for agreement with human judgement. Confusing those two objectives is how this category of product fails.

Done means all eight of these work

Streaming voice loop inside the latency budget • adaptive endpointing tuned against hand-marked boundaries • explicit turn state machine with a degraded state • persona with a question plan and three difficulty tiers • coach with evidence spans and confidence gating • synchronised report replay • human-labelled evaluation set with published agreement • fine-tuned scorer benchmarked against a prompted baseline.
Nothing else is required. The 112-feature catalogue is a roadmap for the README, not a backlog.

2 Architecture

Service	Language	Responsibility
web	TypeScript	Next.js. Server components for history and reports; the practice room is a client island owning capture, the worklet, the socket and playback.
api	Python	FastAPI. Auth, scenario/rubric CRUD, session creation, report retrieval, analytics. Stateless. Never touches audio.
realtime	Python	The latency-critical service. One WebSocket per session, turn state machine, VAD and endpointing, ASR streaming, persona call, synthesis streaming. Stateful — a session is pinned to one process.
coach	Python	Off-path worker. Consumes completed turns from a queue, runs the scorer and the narrative model, writes scores and the report. Its slowness is invisible.
postgres	—	Primary store plus pgvector for scenario similarity and feedback dedup. Neon.
redis	—	ARQ queue, report-ready pub/sub, ephemeral session state, rate limits. Upstash.
s3	—	Audio segments, session recordings, report assets. R2 or Supabase in prod, MinIO locally.

One backend language, because the difficulty is concentrated in a single Python process that must do VAD, ASR, LLM and TTS streaming inside 1.5 seconds. A language boundary there buys nothing and costs a serialisation hop. The contract to the front end is one versioned WebSocket schema generated from a single JSON Schema into both Pydantic models and TypeScript types.

2.1 Tables you need for P0

users • profiles • personas • scenarios • rubrics • rubric_criteria • sessions • turns • turn_metrics • turn_scores • session_scores • reports • latency_events • model_calls • annotations

Skip scenario_versions, model_versions and eval_runs until day 18. Keep **turn_metrics separate from turn_scores** — one is arithmetic on timestamps and is always right, the other is a model judgement and is sometimes wrong; one table invites the UI to present them with equal authority. Keep **latency_events as a table, not a log line** — latency is the headline claim, and a claim you can only support with grep is not an engineering artefact.

2.2 Session lifecycle

Brief compiled and frozen → socket opens with a short-lived token → warm-up (ASR resident, prompt prefix submitted for caching, opening line pre-synthesised) → VAD on every frame → streaming ASR with partial captions → endpointing → utterance finalised as a turn row → persona streams tokens → sentence chunks synthesised incrementally → playback with stop-on-speech → turn pair enqueued to the coach → off-path scoring → close, upload, report, notify.

2.3 The turn state machine

Formalising this as a state machine rather than a tangle of callbacks is what makes real-time behaviour testable. Every transition is logged with a timestamp, which is also how the latency dashboard is built for free.

State	Behaviour	Exits to
idle	Socket open, VAD running on every frame.	listening on onset
listening	Audio buffered, ASR emitting partials, silence timer armed and reset on each voiced frame.	endpointing, aborted
endpointing	Silence threshold crossed; the cascade in §4.2 runs.	thinking, or back to listening
thinking	Utterance finalised, persona streaming. Client shows a thinking indicator, never a spinner.	speaking on first chunk
speaking	Persona audio playing. Microphone still monitored.	idle, interrupted
interrupted	User spoke past the guard interval. Playback stops, the partial turn is recorded as truncated.	listening
closing	Flush, upload, finalise, enqueue the report job.	closed
degraded	A component failed. The session continues reduced — text-only reply or a canned holding line — rather than dying.	any

degraded is a first-class state. Real-time pipelines fail partially, not totally. A TTS timeout should produce a spoken fallback and a logged incident, not a dead session and a lost recording. Modelling degradation explicitly converts the most embarrassing possible demo failure into a barely noticeable hiccup.

3 Stack

Only what P0 touches. Hours are study time; assume you apply it the same day.

Technology	What to learn	Hrs	Status
Transformers + PEFT	Tokenisation, Trainer loop, custom heads, LoRA config, eval callbacks, checkpointing. Skip distributed training.	10	New
faster-whisper	Model sizes and quantisation, the streaming pattern, initial_prompt biasing, word timestamps, real-time factor. Skip fine-tuning Whisper.	8	New
Web Audio + AudioWorklet	Worklet processor lifecycle, resampling, scheduling against AudioContext.currentTime. getUserMedia you already know. Skip effects.	6	Partial
Evaluation methodology	Quadratic weighted kappa, inter-annotator agreement, calibration, split hygiene. Highest interview value per hour on this list.	5	New
Persona prompting	Layered prompts, prefix caching, history compaction, behavioural rules as imperatives.	4	Near
Langfuse	Traces and spans, decorator API, prompt registry. Droppable — first on the cut list.	4	New
SQLAlchemy 2.0	Declarative models, async sessions, relationship loading, Alembic. Skip the legacy query API.	3	New
Docker + Compose	Multi-stage builds, Compose networking, volumes, why model images are large. Skip Kubernetes.	3	Partial
Piper / Kokoro	Voice selection, streaming synthesis, chunk boundaries, sample-rate matching.	3	New
wavesurfer.js	Waveform rendering, regions, playhead synchronisation.	3	New
Stateful deployment	Why realtime cannot be serverless, process sizing, resident model memory, health checks.	3	Partial
Silero VAD	ONNX invocation, frame sizing, probability thresholds, hysteresis. Small surface, high leverage.	2	New
WebSockets, binary	Binary frames, backpressure, heartbeats, reconnection with resume. Socket.IO covers the fundamentals.	2	Near
ARQ	Queues, workers, retries, cron. Tiny surface.	2	New
LiteLLM	Unified completion, routing, fallbacks, streaming, cost callbacks.	2	Near
FastAPI, Pydantic, Postgres, Next.js, React, Tailwind, TypeScript, JWT, Groq, Sentence-Transformers	Already known or trivial extensions of what you know.	—	Known

Roughly fifty-eight hours inside the build window. **Never two Heavy technologies in one day** — AudioWorklet, faster-whisper and Transformers/PEFT are the three, and the day plan schedules them one at a time.

Model routing. Every call resolves a *logical role* from configuration at runtime, never a hard-coded provider: `persona` (fast hosted instruct, selected on time to first token), `persona_local` (Ollama, so the demo survives a dead network), `endpointer` (very small local), `scorer` (fine-tuned local encoder), `narrator` and `planner` (mid-tier hosted, once per session), `judge` (strong hosted, evaluation only, never in the user path), `embedder` (local). This is what makes bring-your-own-key and the offline demo possible.

4 The voice pipeline

4.1 Latency budget

Stage	What happens	p50	p95
Endpoint detection	Silence window elapses. Dominated by a chosen threshold, not by compute.	350 ms	450 ms
ASR finalisation	Decoder flushes; punctuation and timestamps applied. Most work already happened during speech.	100 ms	180 ms
Prompt assembly	History compaction, plan state, cache-marked prefix. Pure CPU.	15 ms	30 ms
Model time to first token	The largest controllable term.	250 ms	400 ms
First chunk assembly	Tokens accumulate to the first clause boundary.	70 ms	120 ms
Synthesis of first chunk	TTS produces the first audio buffer.	150 ms	250 ms
Transport and jitter buffer	Network hop plus the deliberate playback buffer.	100 ms	160 ms
End to end	End of speech to first audible word.	≤ 1100 ms	≤ 1400 ms

Percentiles do not add

The end-to-end p95 target is lower than the sum of the per-stage p95 values, and that is not an error. A run is only at the 95th percentile end to end if several stages are slow simultaneously, which is rarer than any one stage being slow. Measure the end-to-end distribution directly and use per-stage percentiles only to find which component to attack. Being able to explain this is exactly the question a senior engineer asks to see whether the numbers were understood or copied.

The four techniques that buy the budget. (1) *Streaming everywhere* — ASR decodes during speech, the model streams tokens, TTS synthesises per chunk, playback starts on the first buffer; worth more than every other optimisation combined. (2) *Prefix caching* — the static prompt layers are stable for a whole session and marked for provider-side caching. (3) *Short first chunks* — emit the first clause early even when a longer chunk would sound marginally better; the first 200 ms of audio buys more perceived quality than any prosody improvement. (4) *Latency masking* — a pre-synthesised backchannel plays the instant endpointing fires. Use sparingly.

4.2 Endpointing — the highest-leverage twenty lines in the system

Two objectives in direct opposition

Endpointing decides both the largest single term in the latency budget and the most infuriating possible failure — being cut off mid-thought. Every millisecond removed from the silence threshold makes the system faster and more likely to interrupt. Everything else in this project is engineering; this is tuning, and tuning needs a labelled set of real utterance boundaries. **Collect 200 hand-marked boundaries on day 5, not day 19.** When in doubt, bias late: a slightly slow system is tolerable, an interrupting one is not.

The v1 policy is a cascade, cheapest first:

1. **Acoustic silence.** Speech probability below threshold for a continuous window. Base 500 ms.
2. **Adaptive threshold.** The window adjusts per speaker from their own within-utterance pause distribution, clamped to 350–900 ms. Costs nothing; the single best quality improvement available.
3. **Filler suppression.** A transcript tail ending in a hesitation marker extends the window. Hesitations reliably predict continued speech.
4. **Syntactic incompleteness.** A trailing conjunction, preposition or article suppresses endpointing — a regular expression, not a model, and it catches a surprising fraction of cases.
5. **Semantic completeness check.** Only when the window elapsed but the transcript looks unfinished. One binary question to a tiny model, 80 ms budget; on timeout, end the turn.

Two guards around the cascade: a minimum utterance length below which detection is treated as noise, and a maximum turn length after which the persona interrupts — realistic behaviour, not a failure.

5 The two agents

The central architectural idea

The persona and the coach are not two prompts against one model. They are two agents with opposed objectives, on different hardware paths, optimised against different metrics and evaluated by different suites. The persona is optimised for latency and believability and is allowed to be shallow. The coach is optimised for agreement with human raters and is allowed to be slow. Any design that puts scoring logic inside the conversational turn sacrifices both.

Dimension	Persona agent	Coach agent
Path	On the latency critical path	Entirely off it, queue-driven
Objective	Sound like a specific person under time pressure	Agree with expert human raters
Budget	Time to first token under 300 ms	Under 400 ms per criterion, unbounded in practice
Model	Fast hosted instruct, or local fallback	Fine-tuned encoder for scores; hosted model for prose only
Failure mode	Character break, repetition, monologue	Miscalibration, hallucinated evidence
Measured by	Adherence suite and latency percentiles	Quadratic weighted kappa against held-out human labels

5.1 Persona

Three prompt layers. Static (role, hard behavioural rules, reply length cap, safety and character rules, output shape) — identical every call, therefore cacheable. Semi-static (the compiled session brief: scenario, difficulty, temperament, duration, question plan, resume facts, focus areas) — fixed for the session, also cacheable. Dynamic (recent turns verbatim, older turns compacted, elapsed and remaining time, plan position, any pending obligation).

Rules stated as imperatives, because models default to being helpful and helpfulness is precisely the wrong behaviour here: never evaluate, grade, coach or reassure; never reveal that a rubric exists; never break character except under the distress rule; ask one question at a time, since two questions in a turn is the most common realism failure; keep under the word ceiling, because real interviewers speak far less than candidates; follow up on vagueness rather than accepting it politely; never repeat a question already answered.

The question plan is generated once at session start — ordered topics with intended depth and time allocation, held as structured state and updated each turn. A persona without a plan wanders, and worse, produces a session that is not comparable to the user's previous attempt at the same scenario, which destroys the progress tracking the product depends on.

The difficulty ladder must change behaviour, not question wording. Gentle accepts partial answers, hints after a long pause, no interruptions. Standard gives one follow-up on vagueness and keeps to time. Hard gives multiple probing follow-ups, deliberate silences, challenges to claims, interruption of over-long answers, explicit time pressure. A persona that asks harder questions in the same friendly manner is not harder practice.

5.2 Coach

Criteria for all families: structure, specificity, relevance, concision, confidence. Technical and viva add technical depth and trade-off reasoning; negotiation adds anchoring, justification and concession discipline. Each criterion carries an **anchor descriptor for every point on the scale** — the same words are read by the human annotator, the model, and the user in the report. Writing them carefully is the highest-value hour in the entire scoring effort.

Split the number from the prose. Scores come from the fine-tuned model and nothing else. The narrative is generated by a hosted model that is given the already-fixed scores and evidence spans and is forbidden from contradicting them. That separation is what lets the number be measured rigorously while the prose stays fluent — and it means a scorer upgrade improves every historical report without regenerating any text.

Cheap checks first. Pace, filler rate, answer length, speech ratio and question coverage are arithmetic on timestamps. They run before any model is invoked, they are always right, they are free, and they carry a surprising share of the perceived value of the report.

Evidence is verified, never trusted. Every score points at character spans in the turn text. Spans are checked by exact substring match against the transcript; a span that does not appear verbatim is discarded and the score downgraded to low confidence. Hallucinated evidence is the single most damaging failure this product can produce and it is entirely preventable with a substring check.

Confidence gating. Below threshold the product shows *not enough signal*, never a number. A wrong score is far more damaging than a missing one.

6 Cost control

Zero marginal cost is a design constraint, not a hope

Every component has a working free or self-hosted path, and every free tier has a named fallback: Groq for the persona with local Ollama behind it; faster-whisper and Piper local; Colab or Kaggle for the twenty-minute fine-tune; Neon with Supabase behind it; Upstash with a Postgres SKIP LOCKED queue behind it; Hugging Face Spaces for the resident-model services with a Cloudflare Tunnel behind it. Nothing in the architecture requires a paid API.

Audio storage is the only tier that will genuinely bind. Raw 16 kHz PCM is about 1.9 MB per minute; twenty fifteen-minute sessions is roughly 570 MB, which eats a 1 GB tier before half the training set exists. Three measures, in order of impact:

- **Never store persona audio.** It is regenerable from the transcript plus the voice identifier. This alone halves the total.
- **Store Opus, not PCM.** Encode user turns at 24 kbps after transcription — roughly a twelvefold reduction, taking twenty sessions from 570 MB to under 50 MB.

- **Expire by default.** Thirty-day retention with an explicit “keep this one” control for sessions that entered the training set.

Three nested budget ceilings. Tokens per turn, tokens per session, spend per user per month. Breaching the turn ceiling truncates the reply; breaching the session ceiling triggers a graceful wrap-up rather than an abrupt cut; breaching the monthly ceiling blocks new sessions with a clear message.

The demo will be cold-started. Free hosting sleeps, and a recruiter clicking your link at 3 a.m. will wait. Build the model-free /demo sample path — one hour of work, and it is what your link actually needs to survive.

7 The fine-tune — do not cut this

This is the technical centre of the project and the part that is genuinely uncommon. Your resume already claims LLM fine-tuning with no project evidencing it; this section converts a claim into an artefact with a metric, a dataset, a baseline and a chart.

Why fine-tune at all — be able to state all four precisely, which is worth more than the fine-tune itself. *Consistency*: a prompted model returns different scores for the same transcript twice, and progress tracking is meaningless if the instrument moves. *Calibration*: prompted scores cluster mid-scale and drift with wording. *Cost and latency*: five criteria over twenty turns is a hundred scoring calls per session. *The rubric is idiosyncratic*: you wrote the anchors, and modest supervision closes that gap fast.

Task. Given (question, answer transcript, criterion key, anchor descriptors), predict an ordinal score on a five-point scale. v1 is a cross-encoder — DeBERTa-v3-base consuming [question] [SEP] [answer] with one lightweight regression head per criterion. About 180M parameters, trains in under twenty minutes on a free T4, infers in tens of milliseconds on CPU, and produces attribution spans naturally. A LoRA adapter on a 1.5B instruct model is an *ablation row*, not the primary path.

Training configuration. 512 tokens truncated from the answer start; batch 16 with accumulation to 32; LR 2e-5, linear warmup over the first ten per cent, cosine decay; 4 epochs with early stopping on validation kappa, patience 1; dropout 0.1, weight decay 0.01, light augmentation on the training split only. **Three seeds per configuration, always** — kept and averaged at inference, with their standard deviation used as the confidence value shown in the product. Every run logged with the dataset revision hash.

Dataset. Approximately 1,000 user turns, five to seven criteria each, so 5,000–7,000 labels. Three sources deliberately mixed: your own sessions (highest fidelity, lowest volume), ten to fifteen recruited sessions with consent (the most valuable source per hour, and it doubles as user testing), and synthetic answers generated at instructed quality levels to fill the tails that real data never provides. **Split by session, never by turn** — turns from one session share a speaker, nerves and vocabulary, and splitting by turn leaks the speaker across train and test and inflates every metric.

The methodology trap that would invalidate the headline claim

It is tempting to label the training set with a frontier model. If the evaluation set is labelled the same way, you have measured how well a small model imitates a big one — distillation, not agreement with human judgement — and “my fine-tune beats the frontier baseline” becomes incoherent, because the baseline is the ground truth and scores 1.0 by construction.

The rule: **the evaluation set is labelled by humans only and is never shown to any generating model.** Model pre-labelling of the *training* split with human correction is acceptable and should be disclosed with the correction rate. Report distillation agreement and human agreement as two separate numbers. Explaining this distinction unprompted is worth more in an interview than the kappa itself.

Annotation protocol. Annotators see the question, the answer, the audio, the criterion and the full anchors — never the model’s prediction. A 200-turn overlap subset is labelled independently by at least two people, producing the **inter-annotator agreement figure that is the ceiling for every model metric that follows**; disagreements over one point go to adjudication. Order is randomised across criteria and sessions to prevent drift. Compute the ceiling *first*: if humans agree at 0.4, fix the rubric before touching a GPU.

Calibration. Fit isotonic regression per criterion on the validation split, mapping raw output onto the rubric scale. Fitted on validation, evaluated on test, never fitted on test. Report expected calibration error before and after.

Ablation ladder — each row is one line of the results table that makes the project’s central claim: majority class → deterministic features only with a linear model → frontier zero-shot → frontier few-shot with anchor examples (*the real baseline to beat*) → fine-tuned encoder, single seed → three-seed ensemble → ensemble plus calibration (*the shipped configuration*) → LoRA on a small instruct model, for the discussion of why it was not shipped.

The sentence this buys you

“My scorer agrees with expert human raters at 0.71 quadratic weighted kappa against a human ceiling of 0.78, versus 0.52 for a prompted frontier baseline, at a twentieth of the cost and a tenth of the latency — and here is the chart of how that moved across nine training runs and three dataset revisions.” Almost no other candidate can say this. Always quote the ceiling beside the number: without it 0.71 sounds mediocre; with it, it sounds like measurement.

8 Evaluation

Three levels. Conflating them is the most common way a voice-plus-LLM project ends up unable to say anything precise about itself.

Level	Metrics and method
1 — Speech	Word error rate on a fixed fixture set including accented speech and technical vocabulary. Real-time factor per host class (above 1.0 the design fails). Endpoint precision and recall against at least 200 hand-marked boundaries. Time to first audio at p50, p90, p95 — the headline product metric. Synthesis real-time factor.
2 — Persona	Character break rate, judged by a model scanning transcripts for coaching, grading, praise or rubric disclosure; target zero. Question repetition by embedding similarity. Plan adherence. Turn length distribution, where a creeping mean is early warning of prompt drift. Difficulty separation — if gentle and hard are statistically indistinguishable, the ladder is decorative. Safety suite in CI.
3 — Scorer	Quadratic weighted kappa per criterion and overall, always beside the human ceiling. Mean absolute error in rubric points. Spearman, because ranking is what progress tracking depends on. Adjacent accuracy. Expected calibration error. Score stability across repeated runs. Evidence validity, which must be 100% by construction and is measured to prove it. Fairness deltas across speaking rate, accent and vocabulary richness, controlling for human-labelled content quality. Cost and latency per session.

Methodology discipline. Three seeds, report mean and spread — a single run of a non-deterministic system is an anecdote. Hold out a third of the human labels and touch it only when publishing a number. Every figure carries its dataset revision, model version and date; numbers without provenance are marketing. Publish the failures too — a README showing where the scorer is weakest is far more convincing than one showing only the headline.

9 UI — the three surfaces that matter

Everything else is a table or a form you can build in an afternoon. Design rules that carry the impression: calm not clinical, dark-first with a real light mode, one accent colour with green/amber/red reserved exclusively for rubric values, and any element claiming to represent sound driven by real amplitude data — a looping fake waveform is worse than none, because users notice.

9.1 The practice room — /app/practice/[sessionId]

The page the entire project is judged on, and the most restrained design in the specification. The screen is almost empty: a single large persona presence with an amplitude-reactive ring (pulsing while listening, moving with real audio while speaking, distinct restrained state while thinking); one caption line beneath; an unmistakable non-verbal *your turn* indicator; an unobtrusive timer; and mute, end, captions. Nothing else. The eight machine states collapse to four the user perceives: your turn, thinking, speaking, connection trouble.

Non-negotiable behaviours: a visible microphone level meter at all times, because silence must never be ambiguous; the thinking state appears within 200 ms of endpointing, before the reply is ready, since perceived latency is dominated by delay before *any* feedback; upload begins during the session so ending never loses the recording; connection loss shows a calm reconnection state and resumes; Escape does not end the session.

9.2 The report — /app/sessions/[id]

The surface that determines whether users come back. Header with overall score and delta, and actions: practise again, retry one question, share, export, delete recording. Verdict block — three sentences, three strengths, three improvements, each linking into the moment that justifies it, with no generic encouragement. Score panel — one row per criterion with score, confidence, a bar against the user's own history, the anchor text for that value, and contributing turns; insufficient signal reads *not enough signal*, never a number. Delivery panel. The replay — waveform with turn boundaries and score markers and a draggable playhead that moves the transcript, the highlighted turn and the score panel simultaneously. Transcript with evidence spans underlined and click-to-see. A discreet per-turn annotation control, which is how the training set grows without a separate tool.

9.3 Observability and evaluations — /app/observability, /app/evals

End-of-speech to first-audio at p50, p90 and p95 with the target line drawn, at the top of the page, because that is the number the project is about. Stage breakdown stacked so the dominant contributor is obvious. A turn waterfall — one turn expanded stage by stage with prompt, response and audio attached, which is how you find the slow component. Model call table. Cost panel beside the counterfactual figure had every score come from a frontier model, which is the economic argument for the whole training effort. Then the model registry, the per-case grid linking each cell to the turn and its human label, and the regression chart across versions with deployment markers.

10 Twenty-four days

Nothing visual is built until the thing it visualises produces real data, and nothing is trained until there is real data to train on.

Day	Focus	Deliverable
1	Foundation	Monorepo, Compose up with Postgres/Redis/MinIO, full schema and migrations, one JSON Schema generating both Pydantic and TypeScript types.
2	Auth and API	OAuth sign-in, sessions, user and profile model, scoped data access, health endpoints.
3	Content	Three scenario families written properly, four personas, two rubrics with every anchor descriptor. Content, not code — and it blocks the fine-tune.
4	Audio spike	Capture through AudioWorklet, 16 kHz PCM over binary WebSocket, server writes a WAV. Freeze the protocol today.
5	VAD and endpointing	Silero, turn boundary detection, adaptive threshold. Record and hand-mark 200 real boundaries as the tuning set.
6	Recognition	Streaming faster-whisper, partials, finalisation, punctuation, word timings, delivery features.
7	Synthesis	Piper, sentence chunker, chunked streaming to the browser, gapless scheduled playback.
8	Close the loop	First full spoken conversation from a CLI harness. Latency instrumentation written today. Gate day — see below.
9	Orchestration	Formal state machine, timeouts on every state, degraded mode, reconnection with resume, coach enqueue.
10	Persona agent	Layered prompts, question plan, difficulty ladder, character rules, memory compaction, safety rules.
11	Latency work	Prefix caching, chunk tuning, backchannel masking, endpointing tuned against the day-5 set. Hit the budget today or descope.
12	Coach v0	Prompted scorer, deterministic metrics, verified evidence spans, aggregation, narrative report. Start booking day-17 sessions.
13	Practice room	Live session UI, state indicators, level meter, captions, reconnection UX.
14	Report I	Transcript, score panel, delivery panel, waveform rendering.
15	Report II	Synchronised replay, click-to-play evidence, turn inspection, annotation control, retry one question.
16	Shell and onboarding	Dashboard, scenario library, progress page, onboarding wizard with the audio check, settings.
17	Real users	Ten to fifteen recruited sessions with consent. User testing and the best third of the training set. Immovable.
18	Dataset	Synthetic quality-ladder generation for the tails, PII scrubbing, session-level splits, annotation interface.
19	Annotation	Bulk labelling, the 200-turn double-labelled subset, inter-annotator agreement, adjudication.
20	Training	Baselines first, then the fine-tune across three seeds, then calibration. Logged with dataset revision hashes.
21	Evaluation and deploy	Full ablation ladder, regression chart, model registry, scorer promoted to active, evaluations page.
22	Observability	Latency dashboard, turn waterfall, model call table, cost panel, speech suite, nightly CI run.
23	Public surface	Landing page, architecture diagram, README with published metrics, self-host compose, deploy, the model-free /demo path.
24	Demo and buffer	Recorded three-minute walkthrough, published numbers, overrun.

11 Rules

The day 8 gate

By the end of day 8 you should be able to have a spoken conversation with the system from a terminal — no interface, no styling, just audio in, audio out, with per-stage timings printed. If that is not true, do not start the user interface on day 13. Fix the pipeline. A beautiful practice room over a voice loop that stutters, cuts people off, or takes four seconds to reply is the standard way this project fails, and it is obvious to anyone who tries it for thirty seconds.

The day 17 gate

Recruited sessions are the constraint that cannot be compressed. People need scheduling, consent, and a product that works well enough not to waste their time. If day 17 slips, the fine-tune — the most differentiating part of the project — gets squeezed. Ask people on day 12, book them for day 17, over-book by fifty per cent, and treat those bookings as immovable.

1. **A missing score beats a wrong one.** Below the confidence threshold, show *not enough signal*, never a number. The same logic applies to evidence: an unverifiable span is discarded, not shown.
2. **Bias late on endpointing.** A slightly slow system is tolerable; an interrupting one is not. Tune against the day-5 boundary

set, never against feeling.

3. **Never two Heavy technologies in one day.** AudioWorklet, faster-whisper, Transformers/PEFT — and Langfuse if you keep it. Check the plan before you improvise.
4. **Nothing outside P0 before day 13.** The catalogue's 112 features are a roadmap for the README, not a backlog.
5. **Freeze contracts at day 4.** Transport, sample rate, frame size and message schema are load-bearing for everything after. Later changes are schema migrations, because that is what they are.
6. **User speech is untrusted content, never instruction.** Delimit and label it, state in the static layer that the rubric is never disclosed under any framing, block replies containing criterion names, and keep the injection cases in the CI safety suite so a prompt refactor cannot silently reintroduce the hole.
7. **Distress breaks character downward into care.** This product deliberately applies pressure to people who are often already anxious. On any real ambiguity the persona drops character, ends the session and surfaces support. Functional requirement, not a compliance checkbox — and write it into the static layer, test it in the suite, and describe it in the README.
8. **Never claim a fine-tune you did not run.** If Phase 4 is genuinely impossible, ship the prompted scorer labelled as the baseline, publish the human-labelled evaluation set and its agreement figure, and state that the fine-tune is the next milestone with the dataset already collected. That is honest and still impressive. The fabricated number is the one thing here that would actually damage you.
9. **Twenty-four working days is not twenty-four calendar days.** You have a job. Four to five hours on weekdays and a full day each weekend lands this across roughly five calendar weeks. Set the deadline against that honestly at the start.

11.1 Cut in this order if you slip

Langfuse tracing → Kokoro TTS → semantic endpointing → the negotiation scenario family → rewrite coaching and model answers → progress and comparison views → the persona library → backchannel masking.

Never cut: the streaming voice loop, endpointing quality, the turn state machine, the two-agent separation, latency instrumentation, the synchronised report replay, the human-labelled evaluation set, or the fine-tune with its published comparison. Those eight are the project.